

COSC 444 Team Project Progress Journal

Team Members:

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Journal Entry 1: In-Person Meeting – Topic Discussion & Literature Review Task Assignment

Date: January 23, 2025

Location: Lecture hall (first in-person meeting after class)

Attendees: Ethan, Ishika, Mete, Zhenbang Raymond He

Mode: In-person meeting (with concurrent Discord updates)

Discussion & Decisions:

- **Project Topic Selection:**

The team met immediately after class to discuss and decide on a project topic. They agreed to focus on a vehicle-related object detection problem. During the discussion, different ideas were raised (e.g., detecting cars versus trucks/ambulances) to ensure a novel twist on a well-established problem.

- **Literature Review Preparation:**

It was decided that each team member would select five relevant articles (covering both traditional methods such as HOG+SVM and modern approaches) for review at the next meeting.

Action Items:

- Each member is to find and review five articles related to traditional and modern vehicle detection methods.
 - Compile a preliminary list of candidate articles before the next meeting.
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Journal Entry 2: Discussion on Problem Statement & Overleaf Collaboration Setup

Date: February 13, 2025

Attendees: Ethan, Ishika, Zhenbang, Mete

Mode: Discord (with some in-class discussion)

Discussion & Decisions:

- **Problem Statement & Report Structure:**

The team discussed their overall problem statement and agreed on using an extended HOG+SVM framework for multi-class vehicle detection. Two options were debated:

1. Car type classification integrated with detection (e.g., distinguishing SUVs, sedans, trucks)
2. Car detection under occlusion

After weighing the pros and cons, the group leaned toward multi-class detection (option 1) since it is more straightforward to extend a binary classifier to a multi-class scenario.

- **Overleaf Collaboration:**

An Overleaf project has been created for the literature review report. The plan is to divide the report into sections:

1. Abstract
2. Introduction
3. Related Works

(Each member will work on their individual .tex files.)

Zhenbang provided the link and explained the merging process.

Action Items:

- Each member will draft their section (Abstract, Introduction, or Related Works) using the Overleaf project.
- Ensure all images and references are added to the shared folders to simplify merging.
- Continue to refine the problem statement based on feedback from the professor.

Journal Entry 3: Initial Coordination and Literature Review Preparation

Date: February 23, 2025

Attendees: Ethan, Ishika, Zhenbang, Mete

Mode: Discord

Discussion & Decisions:

- **Document Refinements:**
The team focused on updating the main document (*main.tex*) by adding more references, illustrative figures (including a system diagram), and polishing the text and grammar.
- **Evaluation Metrics & Comparative Table:**
Ishika inquired about including evaluation metrics (precision, recall, F1 score) and a table comparing traditional methods with deep learning approaches. Zhenbang confirmed this addition.
- **Reference Coordination:**
Members coordinated to ensure all individual references were integrated into the main file.

Action Items:

- Update *main.tex* with the revised sections, evaluation metrics, and comparative table.
 - Incorporate the shared system diagram link provided by Ethan.
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Journal Entry 4: Final Review & Submission of Literature Review

Date: February 24, 2025

Attendees: Ethan, Ishika, Zhenbang, Mete

Mode: Overleaf collaboration

Discussion & Decisions:

- **Proofreading & Final Edits:**
A final review session was held to proofread the literature review document. The team agreed to stop editing at 11:00 PM.
- **Submission:**
It was decided that one member would submit the report on behalf of the team.

Action Items:

- Confirm submission and send a follow-up email to the TA regarding the next steps.
 - Prepare for the upcoming TA meeting to discuss the proposed project idea and initial data.
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Journal Entry 5: Scheduling the TA Meeting & Preparing the Demo

Date: February 26–27, 2025

Attendees: Ethan, Ishika, Zhenbang, Mete

Mode: Discord

Discussion & Decisions:

- **TA Meeting Scheduling:**

The team coordinated to set up a meeting with the TA to present their proposed project idea. Preferred meeting times were discussed, and Ethan emailed the TA to finalize the slot.

Action Items:

- Finalize the demo environment and ensure the GitHub repository is up-to-date.
 - Develop key talking points for the 10–20 minute TA meeting.
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Journal Entry 6: TA Meeting & Project Demonstration

Date: February 28, 2025

Attendees: Ethan, Ishika, Zhenbang, Mete

Mode: Online (Google Meet)

Discussion & Decisions:

- **Project Demo:**

Zhenbang demonstrated the HOG+SVM-based vehicle detection framework using the GitHub repository. The demo was run on Jupyter Notebook, and the team explained how to set up the environment.

- **Feedback from TA:**

The TA provided positive feedback, indicating that the proposed method was acceptable. He advised that the next major step is adapting the framework into a multi-class classifier and preparing for dataset challenges.

- **Task Delegation:**

The discussion also touched upon upcoming tasks such as dataset collection, data augmentation, and potentially integrating a CNN module to enhance classification accuracy.

Action Items:

- Begin the transition from a binary SVM to a multi-class classifier.
 - Initiate dataset collection and augmentation procedures.
 - Plan for the integration of additional modules if needed.
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Journal Entry 7: Finalizing Project Tasks & Future Roadmap

Date: March 20, 2025

Attendees: Ethan, Ishika, Zhenbang, Mete

Mode: Discord

Discussion & Decisions:

- **Presentation Date Announcement:**
With the final presentation scheduled for April 1, 2025, the team refocused on remaining tasks.
- **Remaining Tasks:**
Key tasks identified include:
 - Multi-class classifier implementation and training.
 - Dataset collection, annotation, and augmentation.
 - Packaging the working demo into an executable format.
 - Final presentation slide preparation.

Action Items:

- Each team member begins work on their assigned module.
 - Schedule follow-up meetings to monitor progress on the remaining tasks.
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Journal Entry 8: Preparation for the project presentation and the final report

Date: March 31, 2025

Attendees: Ethan, Ishika, Zhenbang, Mete

Mode: Discord Audio Meeting

Discussion & Decisions:

- **Presentation Slides**

Team members went through the draft slides together and discussed the further improvements and modifications. More descriptions on introduction and related work should be added and some details on system implementation should be reduced since they are trivial.

- **System setting of the working demo**

Discussed how the working demo should be present and the length and the content of the working demo. Demo stage should be controlled within 3 minutes.

- **Final report revision**

Team members also went through the draft report. Related work occupied too many pages and should be trimmed since this part was already delivered in the literature reviewing report. The training process of the CNN network should be expanded.

Action Items:

- Each team member revises his own part of the slides and reports.
 - Everyone should practice his delivery of the presentation to make sure the presentation time won't exceed the limit.
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